Lecture 2 – Data lifecycle (revisited), curation and more

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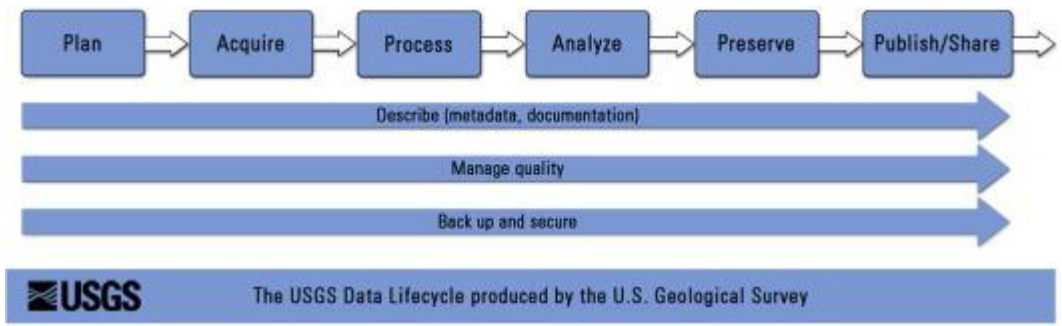
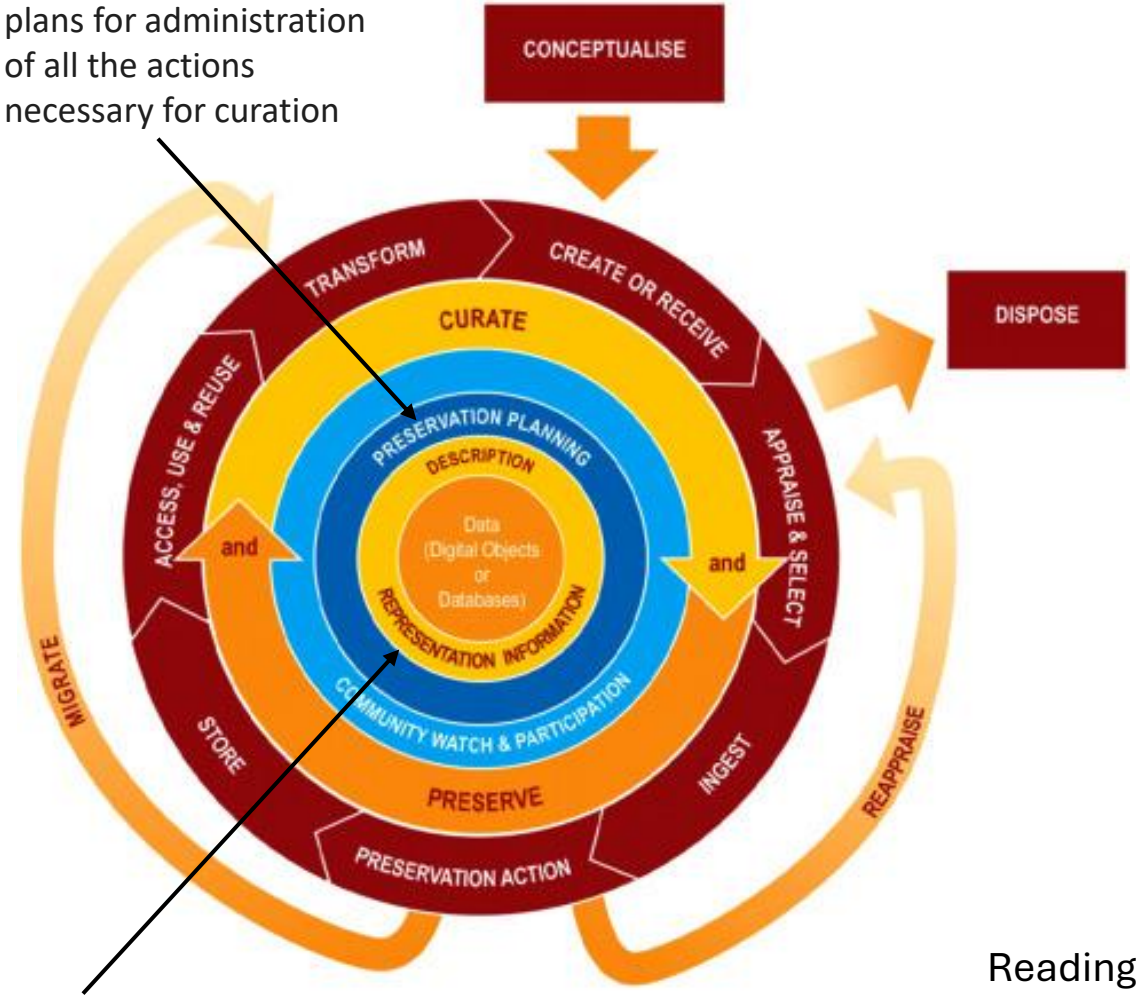
Data lifecycle (revisited)



plans for administration of all the actions necessary for curation



actions of providing metadata

Reading Assignment:
https://www.sciencedirect.com/science/article/pii/B9780128097151000043

Data architectures

Era	Architecture	Key Technologies	Limitations	Business Impact
1960s-1970s	- Hierarchical Databases - Network DBMS	- IBM's IMS - CODASYL	- Rigid structure - Limited flexibility - Complex navigation	- Enabled first digital records - Limited to large enterprises - High maintenance costs
1980s-1990s	- Relational Databases - Data Warehouses	- Oracle - SQL Server - DB2	- Schema rigidity - Scaling challenges - Cost of ownership	- Standardized reporting - Business intelligence - Decision support systems
2000s-2010s	- NoSQL Databases - Big Data Systems	- MongoDB - Hadoop - Cassandra	- Consistency challenges - Complex management - Skill requirements	- Real-time analytics - Unstructured data processing - Scalable operations
2020s-Present	- Data Mesh - Data Fabric - Cloud Native	- Snowflake - Databricks - AWS Services	- Integration complexity - Cost optimization - Governance challenges	- Domain-driven analytics - Self-service data - AI/ML integration

Data architecture components (with example)

```
// Example Data Architecture Layer Definition

1. Data Ingestion Layer
   ├── Batch Processing
   │   ├── ETL Jobs
   │   └── Bulk Data Loads
   └── Stream Processing
       ├── Real-time Events
       └── Change Data Capture

2. Data Storage Layer
   ├── Raw Data Zone
   │   ├── Object Storage (S3)
   │   └── Data Lake
   ├── Processed Data Zone
   │   ├── Data Warehouse
   │   └── Mart Schemas
   └── Serving Layer
       ├── OLAP Cubes
       └── Materialized Views

3. Processing Layer
   ├── Distributed Computing
   │   ├── Spark Clusters
   │   └── Processing Jobs
   └── Query Engines
       ├── SQL Processing
       └── Analytics Engines

4. Consumption Layer
   ├── Analytics Tools
   │   ├── BI Platforms
   │   └── Notebooks
   └── APIs
       ├── GraphQL
       └── REST Services
```

E-commerce Data Platform

```
// Data Ingestion Pipeline Configuration
{
    "pipeline_config": {
        "sources": [
            {
                "type": "clickstream",
                "format": "json",
                "schema_version": "1.0",
                "validation_rules": [
                    {
                        "field": "user_id",
                        "type": "string",
                        "required": true
                    },
                    {
                        "field": "timestamp",
                        "type": "datetime",
                        "format": "ISO8601"
                    },
                    {
                        "field": "event_type",
                        "type": "enum",
                        "values": ["view", "click", "purchase"]
                    }
                ],
                "processing_rules": {
                    "batch_size": 1000,
                    "processing_interval": "5min",
                    "deduplication": true
                }
            },
            {
                "type": "transaction",
                "format": "avro",
                "schema_registry": "http://registry.example.com",
                "validation_rules": [
                    {
                        "field": "transaction_id",
                        "type": "uuid",
                        "required": true
                    },
                    {
                        "field": "amount",
                        "type": "decimal",
                        "precision": 2
                    }
                ]
            }
        ],
        "transformations": [
            {
                "type": "enrichment",
                "lookups": ["user_profile", "product_catalog"],
                "mapping": {
                    "user_id": "profile.details",
                    "product_id": "catalog.metadata"
                }
            },
            {
                "type": "aggregation",
                "windows": ["5min", "1hour", "1day"],
                "metrics": ["count", "sum", "average"]
            }
        ]
    }
}
```

Data challenges/solutions in current architectures

Data Volume Management

- Scale Considerations
- Petabyte-scale data processing
  - Distributed storage systems
  - Partitioning strategies
    - Range partitioning
    - Hash partitioning
    - List partitioning
    - Composite partitioning
  - Compression techniques
    - Column-level compression
    - Table-level compression
    - Custom compression algorithms
- Performance optimization
  - Query optimization
    - Index selection
    - Statistics maintenance
    - Execution plan optimization
  - Caching strategies
    - Result cache
    - Buffer cache
    - Distributed cache

Data Velocity Handling

```
// Stream Processing Configuration Example
{
    "stream_config": {
        "input": {
            "kafka_topics": ["clicks", "transactions", "logs"],
            "partitions": 24,
            "retention_hours": 168
        },
        "processing": {
            "window_types": {
                "tumbling": {"size": "5minutes"},
                "sliding": {
                    "size": "1hour",
                    "slide": "5minutes"
                },
                "session": {"gap": "30minutes"}
            },
            "watermark": "2minutes",
            "late_arrival_handling": "side_output"
        },
        "output": {
            "primary": {
                "type": "elasticsearch",
                "index_pattern": "events-{yyyy.MM.dd}",
                "shards": 5,
                "replicas": 1
            },
            "backup": {
                "type": "s3",
                "bucket": "raw-events",
                "format": "parquet"
            }
        }
    }
}
```

The “it” in AI models is the dataset.

Posted on June 10, 2023 by jbetker

I’ve been at OpenAI for almost a year now. In that time, I’ve trained a **lot** of generative models. More than anyone really has any right to train. As I’ve spent these hours observing the effects of tweaking various model configurations and hyperparameters, one thing that has struck me is the similarities in between all the training runs.

It’s becoming awfully clear to me that these models are truly approximating their datasets to an incredible degree. What that means is not only that they learn what it means to be a dog or a cat, but the interstitial frequencies between distributions that don’t matter, like what photos humans are likely to take or words humans commonly write down.

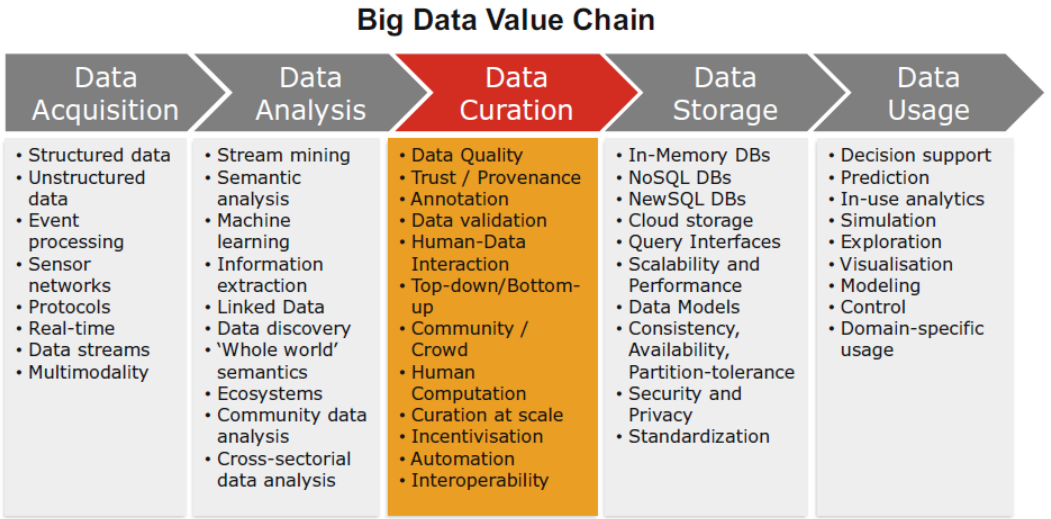
What this manifests as is – trained on the same dataset for long enough, pretty much every model with enough weights and training time converges to the same point. Sufficiently large diffusion conv-unets produce the same images as ViT generators. AR sampling produces the same images as diffusion.

This is a surprising observation! It implies that model behavior is not determined by architecture, hyperparameters, or optimizer choices. It’s determined by your dataset, nothing else. Everything else is a means to an end in efficiently delivery compute to approximating that dataset.

Then, when you refer to “Lambda”, “ChatGPT”, “Bard”, or “Claude” then, it’s not the model weights that you are referring to. It’s the dataset.

Data curation: Definition

- **Definition:** Data curation is the process of organizing, maintaining, and preserving datasets to ensure their accessibility, usability, and quality for analysis.
  - generate FAIR (Findable, Accessible, Interoperable, and Reusable) and analyzable data
- **Core Objectives:**
  - Preserve data integrity.
  - Facilitate accurate data-driven decision-making.
  - Ensure long-term accessibility and compliance.
- **Applications:**
  - Research repositories.
  - Enterprise data management.
  - Big data analytics pipelines.



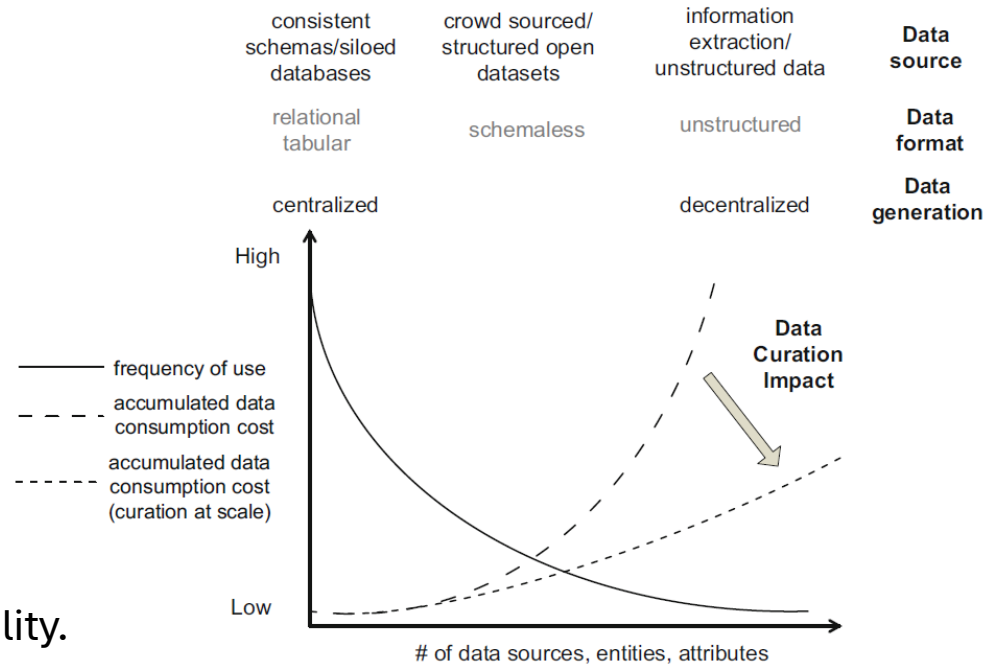
Source: Cavanillas et al., New Horizons for a Data-Driven Economy A Roadmap for Usage and Exploitation of Big Data in Europe

In-class question: What is the difference with data governance?

Data curation: Why?

- **Challenges with Inaccurate Datasets:**
  - Biased: AI systems (e.g., image recognition) may exhibit gender or racial bias.
  - Inaccurate, unreliable, or misrepresented data.
  - Error-ridden or ambiguous datasets.

- **Consequences of Poor Curation:**
  - Lack of processed or curated datasets reduces feature quality.
  - Restricts data development and applications.
- **Benefits of Data Curation:**
  - **Data Quality:**
    - Organizes, describes, cleans, and preserves datasets.
    - Prevents "data swamps" by ensuring proper data storage and segregation.
    - Facilitates long-term accessibility and usability for analysts.
  - **Machine Learning:**
    - Prepares training data for AI/ML by ensuring it is labeled, categorized, and unbiased.
    - Enhances the reliability and machine-readability of datasets.



Source: Cavanillas et al., New Horizons for a Data-Driven Economy A Roadmap for Usage and Exploitation of Big Data in Europe

Source: https://research.aimultiple.com/data-curation/

Data curation: Steps

1. Selection:

1. Identify datasets relevant to the project goals.
2. Evaluate data quality and credibility.

2. Cleaning (next lecture)

1. Handle missing data (e.g., imputation, deletion).
2. Detect and resolve inconsistencies (e.g., duplicate records, incorrect formats).
3. Normalize data for consistency.

3. Organization:

1. Apply schema definitions for structured datasets (e.g., JSON, XML, SQL tables).
2. Use standard naming conventions and folder hierarchies.

4. Enrichment:

1. Add metadata (e.g., author, creation date, format).
2. Integrate data with external sources (e.g., APIs, linked data).

5. Preservation:

1. Store data in durable formats (e.g., Parquet, ORC).
2. Ensure compliance with archival standards (e.g., FAIR principles).

Data curation

	Data Curation	Data Management	Data Cleaning
Definition	Organizing and maintaining data to make it usable and accessible for specific purposes.	The practice of handling all data throughout its lifecycle.	Identifying and removing the inconsistencies and other quality issues from the dataset.
Objective	Ensure data is findable and understandable.	Maintain data integrity, security, and accessibility across the organization.	Prepare data for storage and analysis.
Processes Involved	Data cleaning, transformation, metadata creation, and documentation.	Data governance, security, storage, ingestion, architectural design, and lifecycle management.	Identifying and eliminating duplicate values, correcting data formats, removing outliers, handling missing values, etc., depending on the use case.
Scope	Focuses on specific data assets as per project requirements.	Data management addresses all the data-related aspects of your organization.	Improves data quality for analysis

Data curation: challenges

- **Volume:**
  - Managing petabytes of data in big data environments.
  - Utilizing distributed storage systems (e.g., Hadoop HDFS, Amazon S3).
- **Variability:**
  - Processing structured, semi-structured, and unstructured data.
  - Handling discrepancies in formats and ontologies.
- **Privacy and Security:**
  - Ensuring compliance with regulations like GDPR and HIPAA.
  - Applying anonymization and data masking techniques.
- **Scalability:**
  - Maintaining performance in growing data pipelines.
  - Employing scalable tools like Apache Kafka and Spark.

Data curation using LLM: Example for HR

- Using zero-shot prompting (regular prompt with no example), ask your AI-assistant to remove indication to gender while retaining the meaning

- "She has an excellent experience working for a top-tier consulting firm. This lady is a very good candidate for the job"

Data curation using LLM

- Using few-shot prompting, you need to give a few examples to the LLM to do the following: Change the name of the video name if it includes 1080p in the name and delete if it has 720p. The output should be returned as a JSON response.
- E.g.,: Input IMG_4329_1080p.mp4 should be renamed to IMG_4239.mp4

Removing personally identifiable information (PII)

More in NLP course

```python
class CreditCardDetector(RegexDetector):
    """Remove credit-card numbers from dirty dirty ``text``.

    Supports Visa, MasterCard, American Express, Diners Club and JCB.
    """
    name = 'credit_card'
    filth_cls = CreditCardFilth
    autoload = True

    # Regexes from:
    # http://www.regular-expressions.info/creditcard.html

    # Fake card numbers from:
    # https://www.paypalobjects.com/en_US/vhelp/paypalmanager_help/credit_card_numbers.htm

    # taken from the alphagov fork of scrubadub: https://github.com/alphagov/scrubadub

    # Looking at wikipedia, there are probably more numbers to detect:
    # https://en.wikipedia.org/wiki/Payment_card_number#Issuer_identification_number_.28IIN.29

    # TODO: regex doesn't match if the credit card number has spaces/dashes in

    regex = re.compile((
        r"(?<=\s)"
        r"(?:4[0-9]{12}(?:[0-9]{3})?"            # Visa
        r"|(?:5[1-5][0-9]{2}"                    # MasterCard
        r"|222[1-9]|22[3-9][0-9]|2[3-6][0-9]{2}|27[01][0-9]|2720)[0-9]{12}"
        r"|3[47][0-9]{13}"                       # American Express
        r"|3(?:0[0-5]|[68][0-9])[0-9]{11}"       # Diners Club
        r"|6(?:011|5[0-9]{2})[0-9]{12}"          # Discover
        r"|(?:2131|1800|35\d{3})\d{11})"         # JCB
    ), re.VERBOSE)
```

```python
class TwitterDetector(RegexDetector):
    """Use regular expression magic to remove twitter usernames from dirty
    dirty ``text``.
    """
    filth_cls = TwitterFilth
    name = 'twitter'
    autoload = True

    # https://help.twitter.com/en/managing-your-account/twitter-username-rules#error
    # Twitter user names must be 15 or less charachtors and only contain a-zA-Z0-9_
    # Twitter and admin are not allowed in user names
    # (?<!\w) prevents it matching email addresses
    regex = re.compile((
        r"(?<!\w)@((?!((admin)|(twitter)))[a-z0-9_]){2,15}\b"
    ), re.VERBOSE | re.IGNORECASE)
```

https://github.com/LeapBeyond/scrubadub

Removing PII using LLM

- Try inputting a text into your LLM. Ask the tool to remove any PII

Appraisal in data curation: Exercise

Scenario: A financial institution wants to develop a fraud detection system. They have access to several data sources:

- Transaction Logs (amount, time, location, merchant).
- Customer Profiles (demographics, credit score, transaction history).
- Device Information (IP address, device ID, browser).
- External Fraud Databases (reported fraudulent activities).

- Task: Evaluate these sources based on relevance, quality, provenance, and potential biases.

Appraisal in data curation: Solution

- Transaction Logs: Highly relevant, generally high quality (accurate records), strong provenance (internal system). Potential bias: may not capture new types of fraud.
- Customer Profiles: Relevant (behavioral patterns), variable quality (self-reported data, potential inaccuracies), strong provenance. Potential bias: may reflect existing societal biases.
- Device Information: Relevant (unusual device or location can indicate fraud), variable quality (IP addresses can be spoofed), variable provenance (some data from third-party providers).
- External Fraud Databases: Highly relevant (confirmed fraud cases), variable quality (data may be incomplete or outdated), variable provenance (different sources with varying reliability). Bias: may reflect reporting biases.

Ingestion in data curation: Exercise

- Scenario: A smart manufacturing company monitors its equipment using numerous IoT sensors, generating high-frequency time-series data (temperature, pressure, vibration). They need to build a real-time data pipeline for predictive maintenance.

- Task: Design a data pipeline considering data volume, velocity, format, and real-time processing requirements.

Ingestion in data curation: Solution

- Data Acquisition: Use message queues like Apache Kafka or MQTT to handle the high volume and velocity of sensor data.
- Data Serialization: Use efficient binary serialization formats like Avro or Protocol Buffers to minimize data size and improve transmission speed.
- Real-time Processing: Use stream processing frameworks like Apache Flink or Spark Streaming to perform real-time data cleaning, transformation, and aggregation.
- Data Storage: Store the processed data in a time-series database (TSDB) like InfluxDB or TimescaleDB for efficient querying and analysis.
- Data Visualization and Alerting: Integrate with visualization tools (e.g., Grafana) to monitor equipment health and trigger alerts based on predefined thresholds.

Enrichment in data curation: Exercise

- Scenario: You have a research dataset on customer satisfaction with different product features.
- Task: Create a metadata record using the DataCite Metadata Schema (suitable for research data). In XML and JSON. You can use an LLM for that